Types of Bias in AI

1. **Data Bias**:
    - **Definition**: Arises from imbalanced or non-representative training datasets.
    - **Example**: A facial recognition system trained predominantly on light-skinned individuals may fail to recognize darker-skinned faces accurately.
2. **Algorithmic Bias**:
    - **Definition**: Occurs when algorithms amplify or create disparities based on their design.
    - **Example**: Recommendation engines that reinforce echo chambers by suggesting content similar to users' previous interactions.
3. **Human Bias**:
    - **Definition**: Results from developers' unconscious biases during dataset selection or algorithm design.

Strategies to Mitigate Bias

1. **Diversify Training Data**:
    - **Approach**: Use datasets that reflect the diversity of the target population.
    - **Enhancement**: Incorporate synthetic data to address underrepresented groups.
2. **Bias Audits**:
    - **Objective**: Perform regular audits using fairness metrics such as demographic parity, equalized odds, or disparate impact.
    - **Example Code for Bias Detection in Python**:

```
from sklearn.metrics import confusion_matrix

def evaluate_fairness(predictions, labels, group_attribute): cm = confusion_matrix(labels, predictions)
    fairness_score = cm[1][1] / sum(cm[:, 1]) # Positive predictive value for a specific group return fairness_score
```